

Assignment: Variables and Constants in C

1. Flipkart Product Variables

```
#include <stdio.h>

int main()
{
    char productName[] = "Wireless Earbuds";
    float price = 1499.99;
    double rating = 4.6;

    printf("Product Name: %s (String)\n", productName);
    printf("Price: %.2f (Float)\n", price);
    printf("Rating: %.1lf (Double)\n", rating);

    return 0;
}
```

Sample Output:

Product Name: Wireless Earbuds (String)
Price: 1499.99 (Float)
Rating: 4.6 (Double)

2. Constant Variable for GST Rate

```
#include <stdio.h>

int main()
{
    const float GST_RATE = 18.0;
```

```
float basePrice = 500.0;

float gstAmount = (basePrice * GST_RATE) / 100;
float finalPrice = basePrice + gstAmount;

printf("Base Price: %.2f\n", basePrice);
printf("GST Rate: %.2f%%\n", GST_RATE);
printf("Final Price: %.2f\n", finalPrice);

return 0;
}
```

Sample Output:

```
Base Price: 500.00
GST Rate: 18.00%
Final Price: 590.00
```

Note: The keyword `const` makes the GST rate unchangeable after assignment.

3. Spotify Playlist Information

```
#include <stdio.h>
```

```
int main()
{
    char playlistName[] = "Top Hits";
    int totalSongs = 50;
    float avgDuration = 3.8;
```

```
    printf("My favorite Spotify playlist is %s, it contains %d songs and the average
song duration is %.1f minutes.\n",

        playlistName, totalSongs, avgDuration);

return 0;
}
```

Sample Output:

My favorite Spotify playlist is Top Hits, it contains 50 songs and the average song duration is 3.8 minutes.

4. Correcting Invalid Variable Names

Invalid Variable Name	Correct Variable Name	Reason
1stPlayer	firstPlayer	Variable names cannot start with a number.
\$score	score	Special characters like \$ are not allowed.
total-marks	totalMarks	Hyphens are not allowed in variable names.

Valid Variable Names:

- player_age
- userName

These names already follow standard variable naming rules.

Conclusion

This assignment demonstrated the use of variables, constants, different data types, formatted output, and variable naming rules in the C programming language.